Test Plan

Prompt

Act as a QA Architect and create a test plan for the Inventory Management System (IMS). Include test scope, testing types, environments, test data requirements, key scenarios, and entry/exit criteria. Consider functional requirements, edge cases, and NFRs, including performance, security, scalability, and reliability.

Response

Inventory Management System – Test Plan

1. Test Plan Objective

The objective of this test plan is to validate that the Inventory Management System:

- Meets functional and non-functional requirements
- Maintains accurate inventory tracking
- Supports high-volume product catalogs and concurrent users
- Ensures security, reliability, scalability, and performance across multiple regions

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2. Scope of Testing

In Scope

- Authentication & RBAC
- Product management
- Inventory management
- Stock inward/outward workflows
- Supplier & purchase order management
- Reporting & dashboards
- Alerts & notifications

- Audit logging
- Multi-region and localization support
- API validation
- Performance and scalability testing

Out of Scope

- Third-party ERP integrations (Phase 1)
- Mobile applications
- AI forecasting features

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3. Test Types

Test Type	Objective
Functional Testing	Validate business workflows and features
Integration Testing	Validate interactions between modules/services
System Testing	Validate end-to-end workflows
Regression Testing	Ensure new changes do not break existing features
User Acceptance Testing (UAT)	Validate business usability
Performance Testing	Validate TPS, response time, and scalability
Load Testing	Validate system under expected load
Stress Testing	Identify system breaking points
Security Testing	Validate authentication, authorization, and vulnerabilities
Database Testing	Validate data consistency and integrity
API Testing	Validate REST APIs and error handling
Usability Testing	Validate user experience
Localization Testing	Validate regional/language support
Failover & Recovery Testing	Validate disaster recovery and resilience

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4. Test Environment

Environments Required

- Development
- QA/Test
- UAT
- Staging/Pre-Production
- Production

Infrastructure Requirements

- Multi-region deployment simulation
- Load balancer
- Database cluster
- Monitoring/logging tools

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5. Test Data Requirements

- Product catalog with:
    - 10K products
    - 100K products
    - 500K products
- Multi-warehouse inventory data
- Multiple currencies and locales
- Concurrent transaction datasets
- Historical inventory transactions
- Duplicate/invalid SKU datasets

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6. Functional Test Scenarios

Authentication & User Management

- Valid/invalid login
- Password reset
- Role-based access validation
- Session timeout
- Unauthorized API access

Product Management

- Create/update/delete products
- SKU uniqueness validation
- Barcode validation
- Product search/filter
- Bulk product upload

Inventory Management

- Add/remove stock
- Prevent negative inventory
- Multi-warehouse stock tracking
- Inventory transfers
- Concurrent stock updates
- Stock adjustment approvals

Purchase & Supplier Management

- Create suppliers
- Create purchase orders
- Partial receiving
- Over receiving validation
- PO approval workflow

Dispatch & Returns

- Dispatch inventory
- Return inventory
- Warehouse validation
- Invalid return prevention

Reporting & Dashboard

- Generate reports
- Export reports
- Inventory valuation accuracy
- Large dataset report generation

Audit & Logging

- Inventory audit trails
- User activity logging
- Immutable logs validation

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7. Edge Case Testing

- Duplicate SKU creation
- Simultaneous inventory updates
- Transaction rollback failures
- Backdated inventory transactions
- Bulk upload failures
- Offline/partial sync scenarios
- Expired inventory handling
- Large file imports
- Product archival/deletion with dependencies

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8. Performance & Scalability Testing

Performance Targets

Metric	Target
Concurrent Users	5K–15K
TPS per Region	Up to 500 TPS
API Response Time	<2–3 sec
Search Response Time	<2 sec
Report Generation	<30 sec for large reports
System Availability	99.9% uptime

Performance Test Scenarios

- High concurrent stock updates
- Bulk inventory imports
- Large product searches
- Massive report generation
- Multi-region traffic simulation
- Peak transaction load testing

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9. Security Testing

Areas to Validate

- RBAC enforcement
- SQL injection prevention
- XSS/CSRF protection
- API authentication
- Encryption validation

- Audit log protection
- Session security
- MFA validation

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10. Database Testing

- Data consistency checks
- ACID transaction validation
- Deadlock handling
- Data archival testing
- Index performance validation
- Replication/failover validation

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11. Localization & Multi-Region Testing

- Currency formatting
- Date/timezone handling
- Language support
- Regional tax/rule validations
- Cross-region latency validation

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12. Failover & Disaster Recovery Testing

- Database failover
- Service restart recovery
- Backup restoration
- Region outage simulation
- Data recovery validation

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13. Automation Strategy

Recommended Automation Areas

- Regression test suite
- API testing
- Smoke tests
- Performance scripts
- UI automation for critical flows

Suggested Tools

- Selenium / Playwright
- Postman / Rest Assured
- JMeter / Gatling
- Cypress
- SonarQube
- OWASP ZAP

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14. Entry & Exit Criteria

Entry Criteria

- Requirements finalized
- Stable QA environment available
- Test data prepared
- Build deployed successfully

Exit Criteria

- Critical defects resolved
- High-priority defects closed
- Performance benchmarks achieved
- UAT signoff completed
- Security validation passed

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15. Defect Management

Defect Severity Levels

Severity	Description
Critical	System unusable/data corruption
High	Major feature failure
Medium	Partial functionality issue
Low	UI/minor issue

Defect Workflow

Open → Assigned → In Progress → Fixed → Retest → Closed

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16. Roles & Responsibilities

Role	Responsibility
QA Lead	Overall test strategy
QA Engineers	Test execution
Developers	Defect fixes
Business Users	UAT validation
DevOps Engineer	Environment support
Security Team	Security validation

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17. Risks & Mitigation

Risk	Mitigation
Large data volume issues	Early performance testing
Inventory inconsistencies	Concurrency & rollback testing
Regional latency	Multi-region load testing
Security vulnerabilities	Regular security scans
Unstable environments	Dedicated staging environments

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18. Deliverables

- Test Plan Document
- Test Cases
- Automated Test Scripts
- Test Execution Reports
- Defect Reports
- Performance Test Reports
- Security Test Reports
- UAT Signoff Document